

Technical Design Document (TDD)

Rihla+

Architecture • APIs • Database • Security • Deployment

Version: 1.0

Status: Production Ready (MVP)

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Organization: Numuv Labs

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1. Executive Summary

Purpose

This Technical Design Document describes the architecture and engineering decisions behind **Rihla+**, an AI-powered healthcare companion.

It serves as the technical reference for developers, engineers, and contributors responsible for maintaining and extending the application.

Objectives

- Explain system architecture
- Document database design
- Describe AI integration
- Outline security measures
- Define deployment strategy
- Support future scalability

2. System Overview

Architecture Style

Modern SaaS Architecture

Client

↓

React Application

↓

Supabase Backend

↓

OpenAI Services

↓

Storage & Database

Core Modules

- Authentication
- Dashboard
- Health Profile
- Timeline
- Medical Documents
- AI Assistant
- Appointment Companion
- Insights

- Analytics

3. High-Level Architecture

User



React + TypeScript



Supabase

├─ Authentication

├─ PostgreSQL

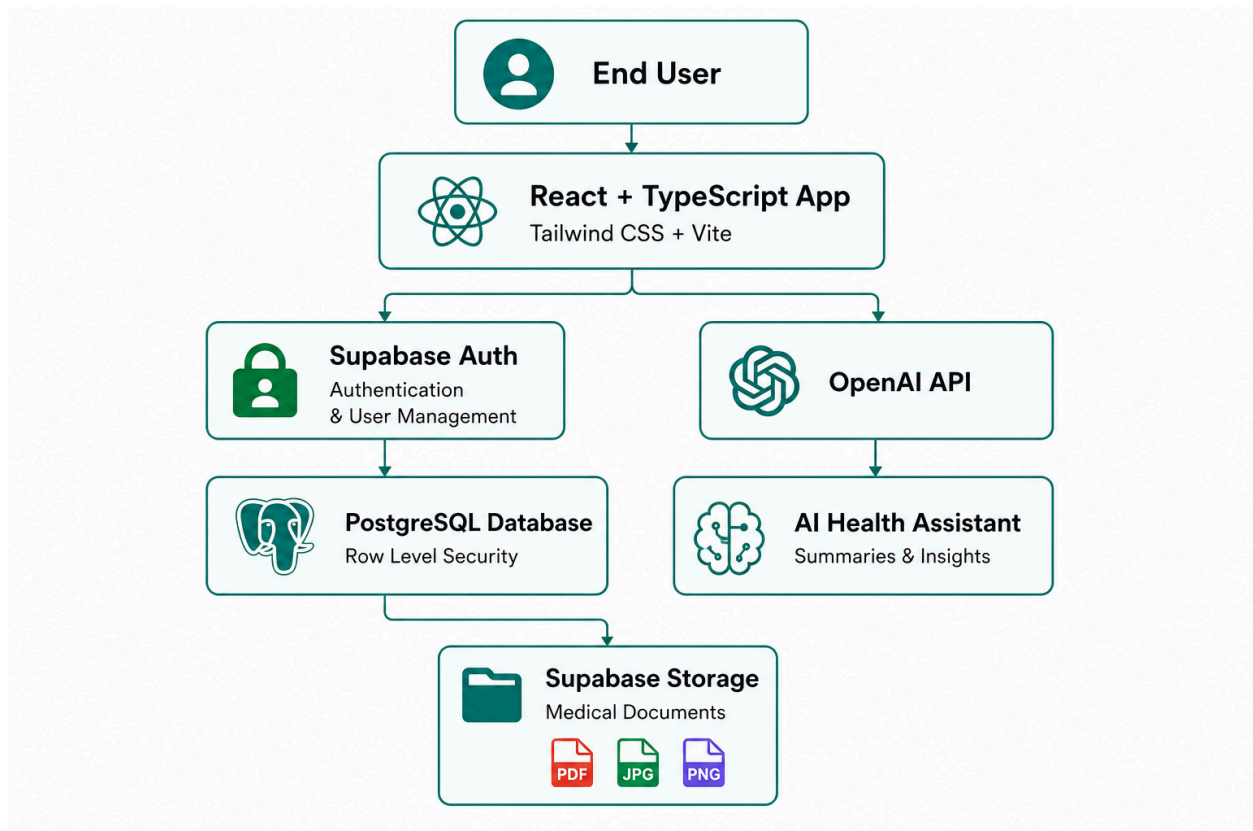
├─ Storage



OpenAI API



AI Responses



4. Technology Stack

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS
- React Router

Backend

- Supabase
- PostgreSQL
- Row Level Security
- Supabase Storage

AI

- OpenAI API

Development

- Lovable
- GitHub
- npm
- ESLint

5. Frontend Architecture

Folder Structure

src/

components/

pages/

hooks/

services/

lib/

types/

utils/

assets/

styles/

Component Philosophy

Reusable

Modular

Composable

Single Responsibility Principle

State Management

- React Context
- React Hooks
- Supabase Client

6. Backend Architecture

Supabase provides:

- Authentication
- PostgreSQL Database
- File Storage
- Row Level Security
- API layer

Business logic remains in the frontend with secure backend services.

7. Database Design

Core Tables

Users

Health Profiles

Symptoms

Appointments

Timeline Events

Documents

Medications

AI Conversations

Insights

Notifications

Relationships

User



Rihla+ — Identity

Supabase (PostgreSQL) · public schema · Row-Level Security on every table

PK

FK / one-to-many

one-to-one



Note: every table has user_id → auth.users(id) ON DELETE CASCADE, enforced by RLS (auth.uid() = user_id).

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Rihla+ · Supabase schema · v1.0

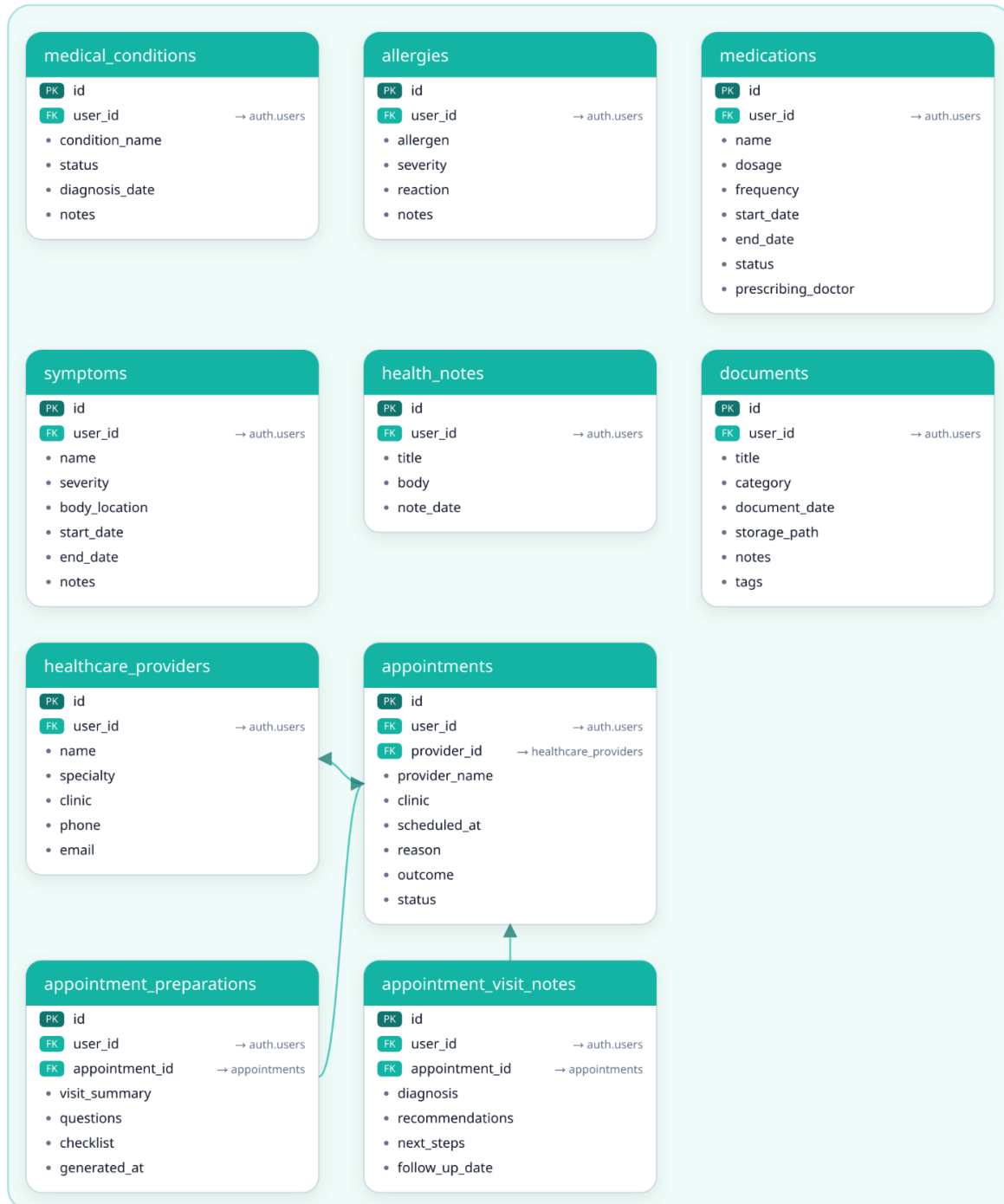
Rihla+ — Clinical

Supabase (PostgreSQL) · public schema · Row-Level Security on every table

PK

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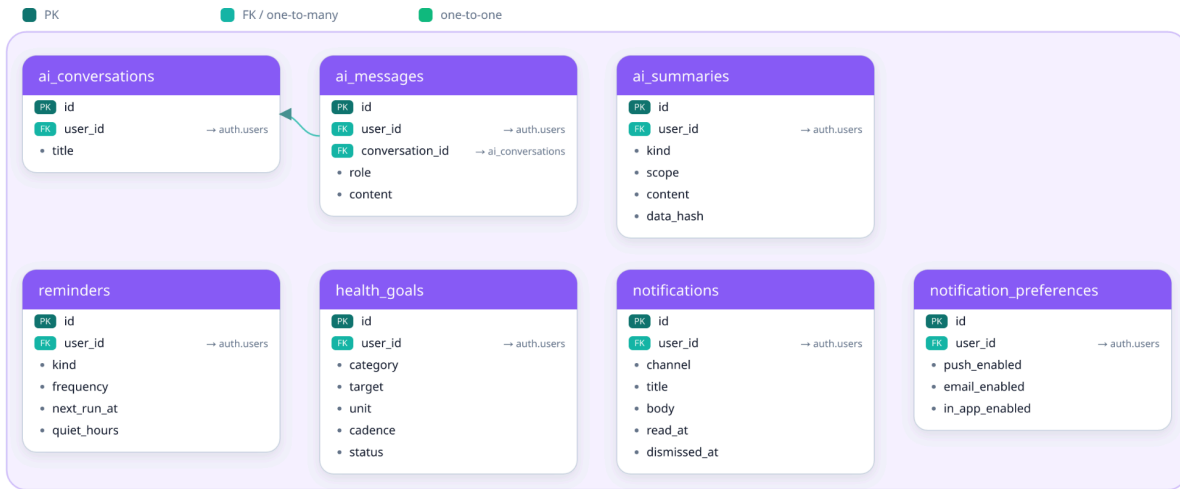
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Rihla+ · Supabase schema · v1.0

Rihla+ — AI & Automation

Supabase (PostgreSQL) · public schema · Row-Level Security on every table



Note: every table has user_id → auth.users(id) ON DELETE CASCADE, enforced by RLS (auth.uid() = user_id).
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Rihla+ · Supabase schema · v1.0

Indexes

- user_id
- created_at
- appointment_date
- timeline_date

8. Authentication

Provider

Supabase Auth

Features

Email Sign Up

Email Verification

Login

Logout

Password Reset

Session Management

Protected Routes

Authorization

Every query uses Row Level Security.

Users can only access their own records.

9. Storage Architecture

Supabase Storage

Buckets

medical-documents/

avatars/

uploads/

Supported Formats

- PDF
- JPG
- PNG

10. AI Integration

Provider

OpenAI

AI Capabilities

Health summaries

Medical explanations

Appointment preparation

Trend observations

Educational insights

AI Workflow

User Question

↓

Health Context

↓

Prompt Builder

↓

OpenAI

↓

Response Validation

↓

Educational Disclaimer

↓

Display

Responsible AI

Every response includes:

Educational only. Not medical advice.

11. API Design

Authentication

POST

/login

/signup

/logout

Timeline

GET

/timeline

POST

/timeline

PUT

/timeline/:id

DELETE

/timeline/:id

Documents

POST

/upload

GET

/documents

DELETE

/documents/:id

AI

POST

/assistant

POST

/appointment-summary

POST

/insights

12. Security

Authentication

Supabase Auth

Authorization

Row Level Security

Storage

Private buckets

Secrets

Environment Variables

OpenAI Keys

Supabase Keys

Validation

Input sanitization

File validation

Authentication middleware

13. Performance

Optimizations

- Lazy Loading
- Code Splitting
- Image Optimization
- Efficient Queries
- Pagination
- Suspense
- Caching

14. Error Handling

User-friendly errors

Retry strategies

Loading states

Empty states

Offline handling (future)

15. Deployment

Frontend

Lovable

Production Build

Backend

Supabase Cloud

Database

Managed PostgreSQL

Storage

Supabase Storage

AI

OpenAI API

CI/CD

GitHub

Automatic Deployments

16. Environment Variables

VITE_SUPABASE_URL=

VITE_SUPABASE_ANON_KEY=

OPENAI_API_KEY=

Never commit secrets.

17. Monitoring

Future

Error Tracking

Performance Monitoring

Analytics

Usage Metrics

Audit Logs

18. Backup & Recovery

Database backups

Storage redundancy

Version history

Disaster recovery procedures

19. Testing Strategy

Unit Testing

Components

Utilities

Hooks

Integration Testing

Authentication

Timeline

AI

Documents

Appointments

End-to-End Testing

Complete user journey

Sign Up

Dashboard

Timeline

AI Assistant

Documents

Insights

20. Future Architecture

Planned

- Mobile Apps
- Apple Health
- Google Fit
- Wearables

- FHIR
- Provider Portal
- Caregiver Sharing
- Offline Sync

21. Engineering Principles

- Modular Architecture
- Security by Design
- Privacy First
- Accessibility
- Reusability
- Scalability
- Performance
- Responsible AI

22. Appendix

Repository

GitHub Repository

(Add Link)

Documentation

PRD

README

Architecture

ERD

Case Study

Tech Stack

React

TypeScript

Tailwind CSS

Supabase

PostgreSQL

OpenAI

Lovable